

The linear power spectrum of the model-independent analysis what is fixed, what is free (DESI DR1 FS+BAO, model Ag245F)

4 October 2026

1 The model

$$P_{\text{lin}}(k, z_i) = \underbrace{\left[\frac{D(z_i)}{D(z_N)} \right]^2}_{\text{growth}} \underbrace{a(k)}_{\text{broadband}} \underbrace{T_{\text{nw}}(k)}_{\text{smooth fiducial}} \left[1 + \underbrace{(1 + A)}_{\text{BAO amplitude}} \underbrace{e^{-k^2 \Delta \Sigma^2 / 2}}_{\text{damping}} \underbrace{O(\alpha_{r_s} k)}_{\text{fiducial wiggles, rescaled by } r_s} \right] \quad (1)$$

Blue = free, sampled. Black = fixed, computed once from the fiducial cosmology and never changed.

Check. At the fiducial values $a \equiv 1$, $A = 0$, $\Delta \Sigma^2 = 0$, $\alpha_{r_s} = 1$ and $z_i = z_N$, eq. (1) gives $P_{\text{lin}}(k, z_N) = T_{\text{nw}}(k) [1 + O(k)] = T(k)$, the fiducial spectrum, exactly.

Redshifts. The spectrum is reconstructed at a single redshift, $z_N = 1.491$ (QSO), the farthest of the six. At the other sample redshifts, $z_i \in \{0.295, 0.510, 0.706, 0.919, 1.317\}$, it has the same shape and only the amplitude changes, through $D(z_i)/D(z_N)$.

2 The fixed part: the fiducial model

- **Fiducial cosmology** (the DESI fiducial, Planck 2018): flat Λ CDM with $\omega_b = 0.02237$, $\omega_{\text{cdm}} = 0.1200$, $h = 0.6736$, $\ln(10^{10} A_s) = 3.044$, $n_s = 0.9649$.
- $T(k) \equiv P_{\text{lin}}^{\text{fid}}(k, z_N)$: its linear spectrum at z_N . It is computed with CosmoPower at $z = 5$ and grown to z_N with Λ CDM growth.
- **The split** $T = T_{\text{nw}} (1 + O)$, done once:
 - $T_{\text{nw}}(k)$ is the smooth part: the Eisenstein–Hu no-wiggle shape times the ratio T/T_{EH} smoothed with a Gaussian of 0.25 dex in k .
 - $O(k) = T/T_{\text{nw}} - 1$ is the BAO wiggle pattern. It is at most 6.6%, its largest peak is at $k = 0.050 \text{ Mpc}^{-1}$, and it is set to zero below $3 \times 10^{-3} \text{ Mpc}^{-1}$.
- k is in Mpc^{-1} , converted with the fiducial h .

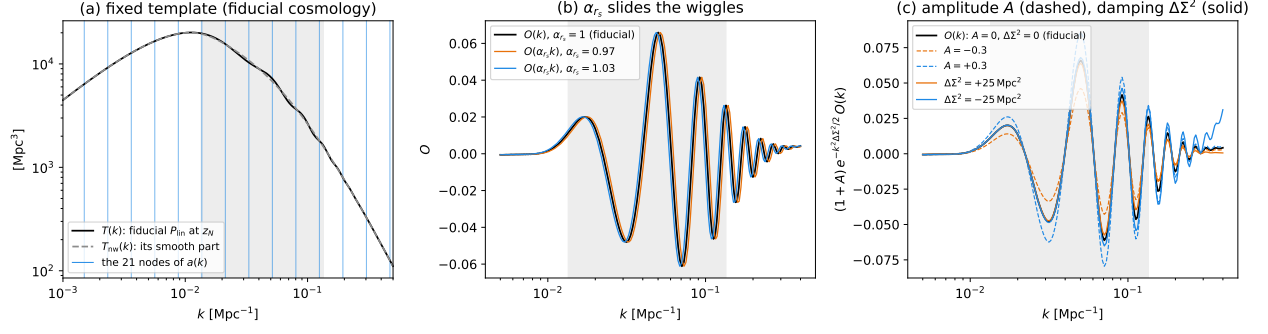


Figure 1: (a) The fixed template T , its smooth part T_{nw} , and the 21 nodes of $a(k)$; the grey band is the data range, 0.02 – $0.2 h/\text{Mpc}$. (b) α_{r_s} slides the wiggles. (c) A scales them at every k (dashed); $\Delta\Sigma^2$ damps them, or undamps them when negative, more strongly at high k (solid).

3 The free parameters

parameter	number	what it does	prior
$D(z_i)/D(z_N)$	5	Amplitude at sample redshift z_i relative to z_N (it is 1 at z_N).	amplitude at each z_i : ± 0.6 in $\ln P$
$a(k)$	21	Broadband shape and the overall amplitude at z_N . $a = \exp[\text{cubic spline of } \ln a_j \text{ in } \ln k]$, with nodes uniform in $\ln k$ over 10^{-4} – $0.7 h/\text{Mpc}$ (spacing 0.443), constant outside.	shape: ± 0.5 per node, plus smoothness; mean of $\ln a_j$: amplitude prior
α_{r_s}	1	Rescales the wiggles by the sound horizon: a peak at k_p moves to k_p/α_{r_s} . In a cosmology, $\alpha_{r_s} = (h r_d)/(h r_d)_{\text{fid}}$.	$\ln \alpha_{r_s} \sim \mathcal{N}(0, 0.05)$
A	1	BAO amplitude: the wiggles are multiplied by $(1 + A)$ at every k .	$\mathcal{N}(0, 0.3)$
$\Delta\Sigma^2$	1	Damping of the linear wiggles relative to the fiducial, $e^{-k^2\Delta\Sigma^2/2}$; negative means less damped.	$\mathcal{N}(0, 25 \text{ Mpc}^2)$
total in P_{lin}	29		

Priors in detail.

- The shape prior acts on $\ln a_j - \langle \ln a \rangle$. It is $\sigma = 0.5$ per node plus $\lambda \sum_j (\ln a_{j+1} - 2 \ln a_j + \ln a_{j-1})^2$, with $\lambda = 7.78$.
- The amplitudes $u_i = \langle \ln a \rangle + 2 \ln[D(z_i)/D(z_N)]$, with $u_N = \langle \ln a \rangle$, are Gaussian with covariance $(0.5^2/21) \mathbf{1}\mathbf{1}^T + 0.6^2 \mathbb{K}$.
- All priors are centred on the fiducial.

4 Relation to a standard BAO template

A Fourier-space BAO fit models the observed spectrum as $P_{\text{nw}}(k) [1 + O(k') e^{-k^2 \Sigma^2/2}]$, plus a free broadband. k' is k dilated by $\alpha = (D/r_d)/(D/r_d)_{\text{fid}}$. Eq. (1) has the same structure: a fixed fiducial wiggle pattern, rescaled only through the sound horizon, with a free amplitude and damping. It differs in three places.

1. **It is the *linear* spectrum.** It goes through the one-loop EFT, so the nonlinear BAO damping comes from pybird’s IR resummation. $\Delta \Sigma^2$ is only the change of the *linear* (Silk) damping from the fiducial, which depends on ω_b/ω_m . It is small and can be negative.
2. **The broadband multiplies the wiggles.** $a(k)$ is free but too coarse (21 nodes) to follow or slide a wiggle, so it carries no BAO information. Because it multiplies the wiggles too, it cannot change their size relative to the smooth part; a BAO fit gets that freedom from its additive broadband terms. A provides it here. Without A (damping alone) real spectra cannot be represented.
3. **The BAO dilation is split in two.** α_{r_s} rescales the wiggles; the per-redshift AP dilations $\alpha_{\parallel}(z_i)$ and $\alpha_{\perp}(z_i)$ dilate the whole spectrum, wiggles and broadband alike. The BAO points depend only on the combination $\alpha/\alpha_{r_s} = (D/r_d)/(D/r_d)_{\text{fid}}$, as in a BAO fit. α_{r_s} itself is close to prior-limited.

Not part of P_{lin} : the growth rate $f(z_i)$ and the AP dilations $\alpha_{\parallel}(z_i)$ and $\alpha_{\perp}(z_i)$ enter afterwards, in pybird’s redshift-space and AP mapping.